



Research on Screening of the Habitability of Exoplanets

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Background

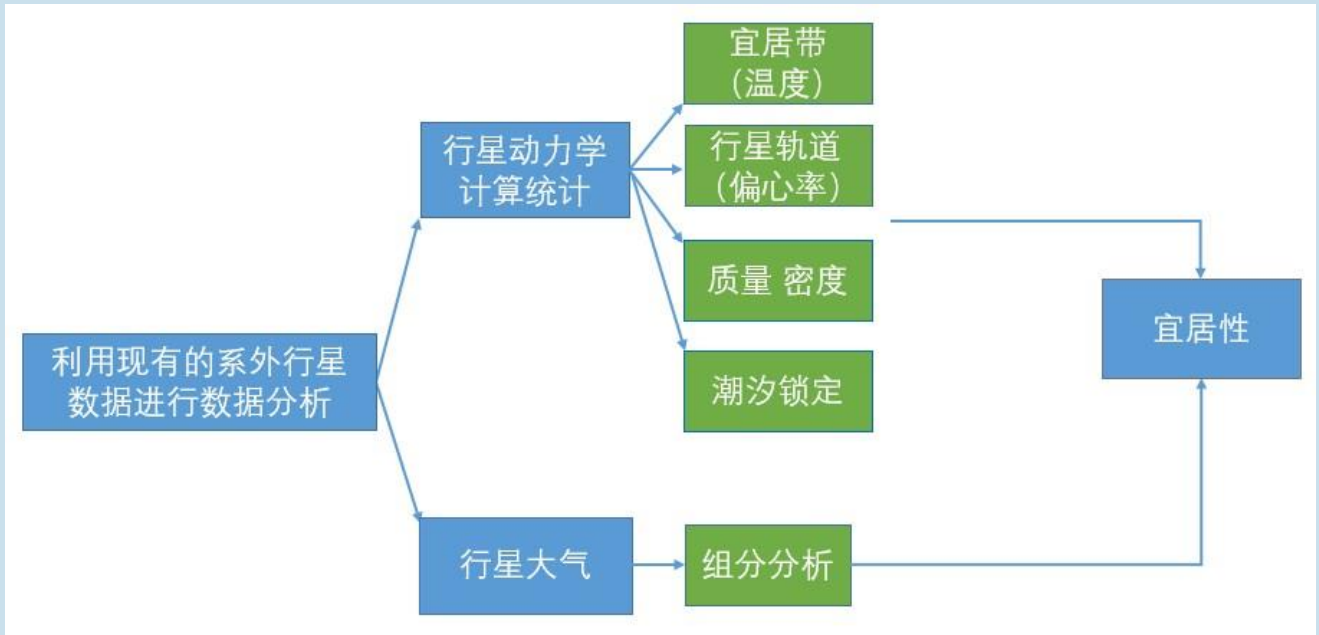
By 2023, a total of 5506 exoplanets had been discovered and confirmed, providing a substantial dataset for the analysis of habitability and potentially habitable planets. Research purpose: To conduct research on existing exoplanet observation data, hoping to find out exoplanets that highly meet the requirements for human habitation.



A habitable planet must have a solid surface and a constant temperature to ensure the existence of liquid water. So we need to establish criteria based on mass, volume, diameter, period, orbital eccentricity, and inclination of the planets in the system. Also, to exclude tidally locked planets.

Research Process

- Step1: Obtain data. Source: <https://www.exoplanet.eu/catalog/>
Step2: Analysis on habitable range, mass, density, eccentricity & tidal locking, to screen habitable planets.
Step3: Analyze and eliminate error



Research Method

1.Habitable zone

$$S_{eff} = S_{eff\odot} + aT_*^2 + bT_*^3 + dT_*^4 \quad T_* = T_{eff} - 5780K \quad d = \left(\frac{L/L_\odot}{S_{eff}}\right)^{0.5}$$

- 2.Density, 0.2ME-10ME is selected as the habitable range; based on the exploration data of terrestrial planets in the solar system, the solid material density of 3g/cm3 is selected as the lower limit.
3. Eccentricity of a habitable planet should be as close to 0 as possible. Planets with eccentricities lower than 0.1 are selected as habitable planets.

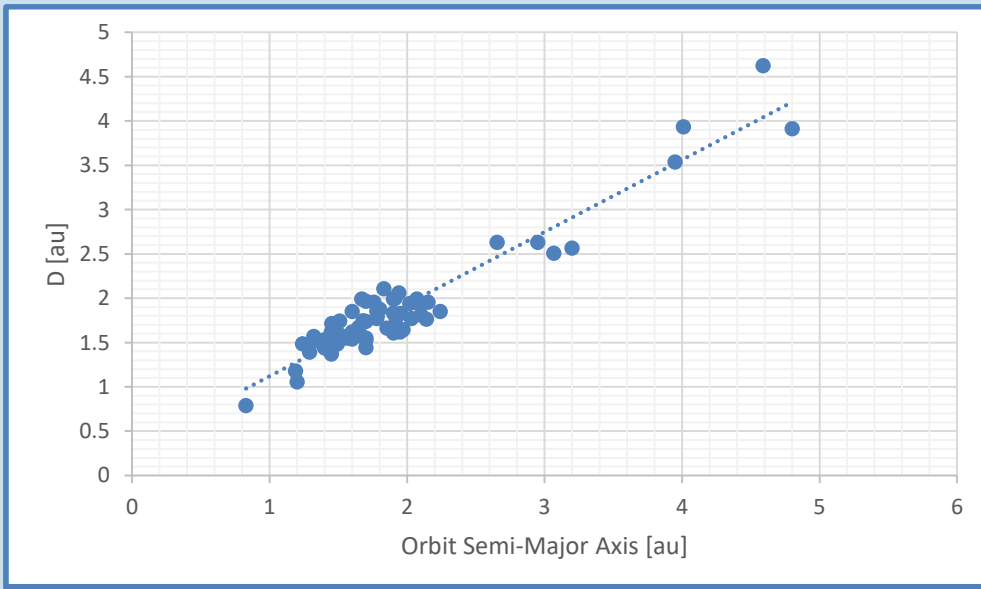
4. Tidal locking

$$\tau_{sync} \approx \frac{4}{9} \alpha Q_p' \left(\frac{R_p^3}{GM_p}\right) (\omega_i - \omega_f) \left(\frac{M_p}{M_*}\right)^2 \left(\frac{d}{R_p}\right)^6$$

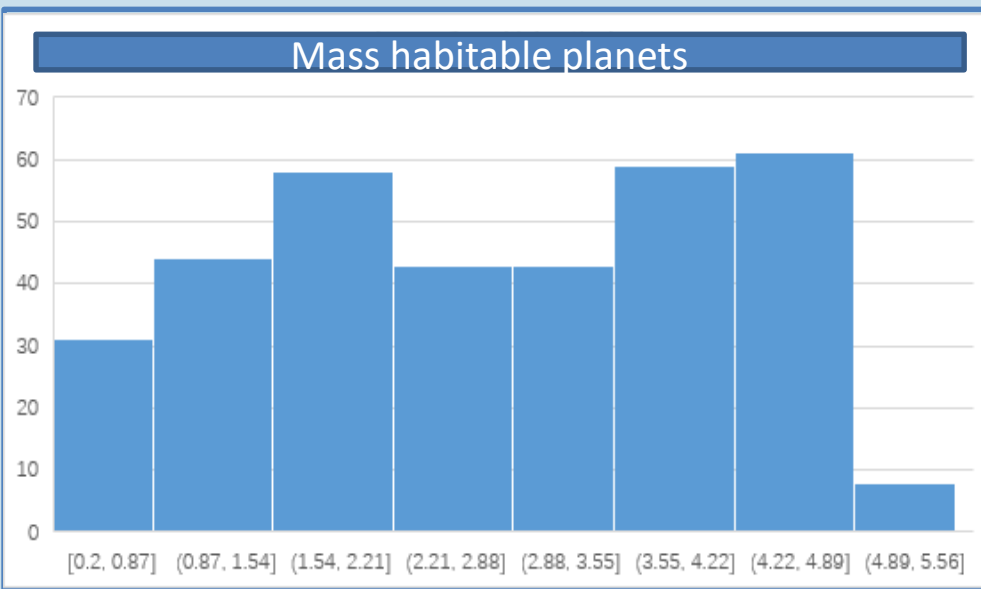
5. Analyze and eliminate error: 1) Explore the correlation between calculated results and measured results to ensure that the calculations are correct. 2) Use detection data from Venus and Mars to verify the accuracy of the model and ensure the screening method is rigorous.
6. Atmospheric components
atmospheric model: <https://psg.gsfc.nasa.gov/atmosphere.php>.

Results

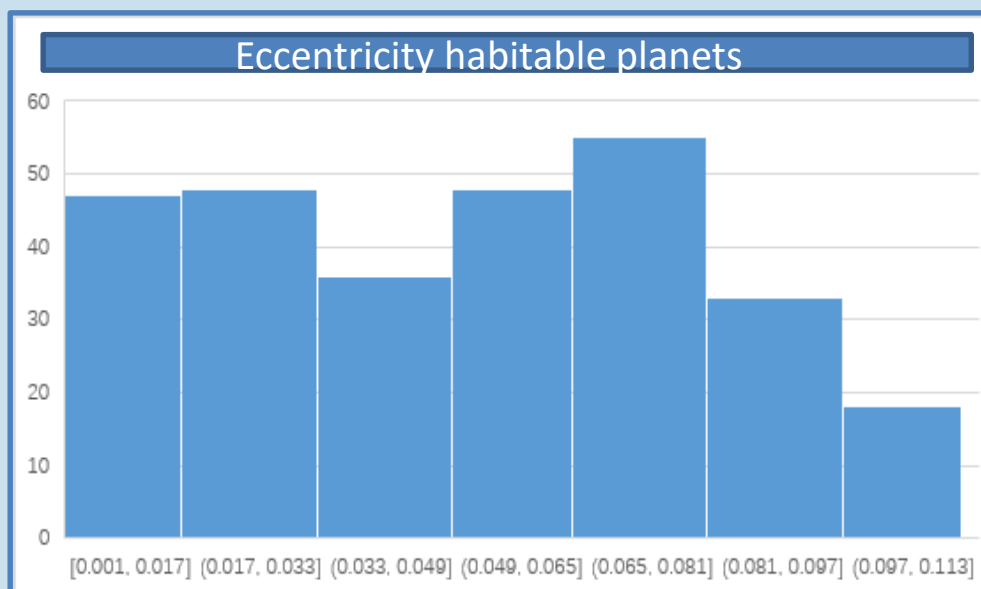
Among the livable exoplanets, there are 65 habitable targets with “Modern Venus”, 83 habitable targets with “Runaway Greenhouse”, 63 habitable targets with “Moist Greenhouse”, 15 habitable targets with “Maximum Greenhouse” and 15 habitable targets with “Early Mars”. 420 mass habitable targets; 119 habitable targets with calculated density, and 962 habitable targets with observed; 1453 habitable targets with eccentricity; 301 habitable targets in tidal locking.
The accuracy of the model has been verified. When tested using Venus and Mars, the results reflected by the model are consistent with the actual detection situation.



Habitable zone (part)



mass



eccentricity

Not tidally locked exoplanets				
AU Mic b	COCONUTS-2 b	CoRoT-4 b	EPIC 229004835 b	GU Psc b
AU Mic c	CoRoT-10 b	CoRoT-6 b	EPIC 249893012 c	HAT-P-15 b
BD+20 594 b	CoRoT-20 b	CoRoT-9 b	EPIC 249893012 d	HAT-P-2 b
BD+60 1417 b	CoRoT-3 b	EPIC 211945201 b	GJ 143 b	HATS-17 b
CFHTWIR-1 b	CoRoT-30 b	EPIC 220674823 c	GQ Lup b	HD 106315 b
Oph 98 b				

Tidal locking (part)

Discussion

The result differs from the research hypothesis of "highly habitable terrestrial exoplanets with dynamic properties exist." Based on the systematic theoretical calculations of this project, previous assumptions and expectations for habitable terrestrial exoplanets have been re-examined from theoretical calculations.
The computational screening of this project has strict requirements for habitable conditions such as habitable zone, density, eccentricity, and tidal locking. In addition, most of the exoplanets discovered at present are large Gas Giant planets. So there are few terrestrial planets to be selected. Now there is no exoplanet as perfect as Earth. Exoplanets with strict habitable conditions in the inner solar system may not have been discovered yet. And I will continue to improve my model in the future.

Innovations

- This project completed the habitability screening of important factors for exoplanets and drew conclusions about the of habitability theory.
- It further raises doubts about the existing habitable theory of exoplanets.
- Constructed LY Habitability Model, which analyzes exoplanet habitability systematically.

Further Research

- The types of exoplanets can be carefully divided according to their properties, not only considering the habitability of terrestrial planets.
- Referring to moons with atmospheres in solar system, continuing to explore the habitability of exomoons, and further supplementing the habitability model.
- After the completion of the large-scale optical space telescope CSST, a new batch of directional observation plans will be launched and a more complete database will be established.